

KAVACH — Incident & Prevention Report

Gas Release Incident Replay — Coke Oven Battery 4

Synthetic composite. This report is generated from a deterministic digital-twin simulation. The scenario is a fictional composite inspired by publicly reported investigations of coke-oven gas accidents in Indian heavy industry. All identifiers, readings, dates and times are synthetic. No real person, plant or event is depicted.

Scenario: vizag_replay · **Horizon:** 600 min from 02:00 · **Generated by:** KAVACH deterministic engine (v-report-1)

1. Executive summary

KAVACH raised a **CRITICAL** compound-risk alert in **Battery 4 Basement / Inspection Chamber** at **06:30** (t=270 min), driven by rules [R1, R3, R5, R4] at score 100/100 and confidence 0.91.

A conventional single-sensor alarm system would not have demanded attention until **09:39** (t=459), when GD-CO4-203 first held an uncleared alarm.

Metric	KAVACH (compound)	Single-sensor baseline
First actionable detection	06:30 (t=270)	09:39 (t=459)
Lead time over baseline	189 min (3.15 h)	—
Warning before incident	240 min	51 min
False negatives (missed hazards)	0	0
False positives	0	0

KAVACH raised a compound alert 189 min (3.1 h) before the first single-sensor alarm, with 0 missed hazard(s) vs baseline's 0.

2. How the hazard compounded

Each factor below was individually explainable and stayed below — or was acknowledged away from — single-sensor alarm levels. Their *co-occurrence* is the hazard a threshold system structurally cannot see:

1. Sub-threshold gas-main pressure drift from 02:30 (regulator PR-104 passing)
2. Transient pressure excursions acknowledged without trend review (04:25, 04:55)
3. Incomplete shift handover at 06:00 — drift not communicated
4. Single-point gas test for confined space entry (chamber rear untested)
5. Confined space entry without gas-main isolation at 06:30
6. Hot work permit issued 20 m from basement vent shaft at 08:30
7. Valve V-707 throttling at 09:30 accelerating back-pressure

3. KAVACH detection and evidence

Alert KV-cob4_basement-270 — 06:30 — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R4] — score 100/100

Rules fired:

- **R1 Gas trend in occupied confined space** (high): CSE-2093 active (crew of 5); gas-main drift on PT-GM-104
- **R3 Confined entry without gas-main isolation** (high): CSE-2093 isolations omit gas-main; network abnormal (PT-GM-104)
- **R5 Single-point gas test** (escalator): CSE-2093 cleared on a single-point gas test; zone trend now rising
- **R4 Shift-handover blindspot** (escalator): shift changeover with unacknowledged drift on PT-GM-104

Key readings at detection:

- PT-GM-104: 8.6 kPa — sub-threshold drift

Regulatory basis for this alert:

- **FA-1948, S.36(2)** — No confined-space entry without gas removal, competent-person certificate or breathing apparatus. No person shall enter such a confined space until all practicable measures have been taken to remove gas, fume, vapour or dust and bring it within permissible limits, and unless either (a) a competent person has certified in writing, on the basis of a test carried out by himself, that the space is reasonably free from dangerous gas/fume/vapour/dust, or (b) the person wears suitable breathing apparatus and a belt securely attached to a rope whose free end is held by a person outside the space.
- **OISD-STD-105, Sec. Confined Space / Vessel Entry Permit** — Positive isolation, purging, gas testing and standby before entry. Entry into a confined space or vessel is permitted only after the equipment is positively isolated from all connected sources (blinding / spading / double-block-and-bleed), drained and purged, and gas-tested (hydrocarbons in %LEL, toxic gases in ppm, oxygen in %). A trained standby person shall be posted at the entry, and gas monitoring shall continue for the duration of the entry.
- **DGMS, CMR-2017 (gas testing by competent persons)** — Gas testing before and during work by a certified competent person. Testing for inflammable and noxious gases in gas-liable workings shall be carried out before work begins and repeated during the work by a certified competent gas-testing person using approved detectors. A single check at one location and one time is inadequate where the atmosphere can change as work proceeds or as connected pressures shift.
- **DGMS, DGMS continuous monitoring & auto-trip** — Continuous CH₄/CO monitoring with automatic withdrawal thresholds. In gassy workings, continuous tele-monitoring of methane (CH₄) and carbon monoxide (CO) shall be provided, with automatic cut-off of power and withdrawal of persons when CH₄ reaches about 0.75% or noxious gas such as CO reaches about 50 ppm. The threshold response is a last-line trip: it reacts at the alarm limit, not to the trend building toward it.

Escalation cascade

- **08:30 (t=390): + [R1, R3, R5, R2] — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R2] — score 100/100**
- **09:31 (t=451): + [R1, R3, R5, R2, R6] — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R2, R6] — score 100/100**

- **10:46** (t=526): + [R1, R3, R5, R2, R7] — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R2, R7] — score 100/100
- **10:53** (t=533): + [R1, R3, R5, R2, R7] — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R2, R7] — score 100/100
- **11:05** (t=545): + [R1, R3, R5, R2, R7] — CRITICAL compound risk in Battery 4 Basement / Inspection Chamber [R1, R3, R5, R2, R7] — score 100/100

4. What the single-sensor baseline did

The baseline's first uncleared alarm was **GD-CO4-203** at **09:39** (t=459) — 189 minutes after KAVACH, and only 51 minutes before the incident. A threshold system reacts to a line being crossed; it cannot see the hours-long convergence that crossed it.

5. Emergency response plan (orchestrator)

On the critical alert, KAVACH's deterministic orchestrator issues:

1. [T+0] Declare **COMPOUND-CRITICAL** — **Battery 4 Basement / Inspection Chamber**

KAVACH score 100/100 at 06:30 from rules [R1, R3, R5, R4] (confidence 0.91). Broadcast control-room alarm; freeze the area.

Basis: DGMS, DGMS continuous monitoring & auto-trip, FA-1948, S.36(1), FA-1948, S.36(2)

2. [T+0] **STOP WORK** — **suspend active permits in the hazard network**

Suspend: CSE-2093 (Confined Space Entry, crew 5) Block issuance of any new hot-work / entry permits in Battery 4 Basement / Inspection Chamber and connected zones until cleared.

Basis: DGMS, DGMS (Tech.) Circular No. 4 of 2020, FA-1948, S.36(2), FA-1948, S.36A

3. [T+2 min] **Positively isolate the Battery-4 riser**

Close AND blind isolation valve V-707 (ZI-GM-707) at the riser; do NOT merely throttle — throttling raises back-pressure on the PT-GM-104 leg. Reduce the isolated section to atmospheric pressure before any intervention.

Basis: DGMS, DGMS (Tech.) Circular No. 4 of 2020, FA-1948, S.37(3), OISD-STD-105, Sec. Isolation

4. [T+2 min] **Notify and mobilise response roles**

Escalate to: Shift Supervisor (S. K. Murthy), Safety Officer, Control Room, Fire & Gas Rescue.

Basis: OISD-STD-105, Sec. Work Permit System (General)

5. [T+5 min] **Evacuate confined space; clear connected zones**

Withdraw and account for the crew in Battery 4 Basement / Inspection Chamber (confined space, no re-entry). Clear connected zones to the muster point and confirm headcount.

Basis: DGMS, DGMS continuous monitoring & auto-trip, FA-1948, S.36(1)

6. [before any re-entry] **Re-test the chamber rear and riser side**

The entry gas test was single-point at the chamber entrance. Re-test the rear/riser side with a representative, competent-person test before considering re-entry.

Basis: DGMS, CMR-2017 (gas testing by competent persons), FA-1948, S.36(2), OISD-STD-105, Sec. Confined Space / Vessel Entry Permit

7. [continuous] **Continuous CH₄/CO monitoring until the trend reverses**

Establish continuous monitoring on the Battery-4 network and hold all work until pressure and CO trends reverse and isolation is verified positive.

Basis: DGMS, CMR-2017 (gas testing by competent persons), DGMS, DGMS continuous monitoring & auto-trip, OISD-STD-105, Sec. Confined Space / Vessel Entry Permit

Evacuation (from the plant connectivity graph):

- Immediate: Battery 4 Basement / Inspection Chamber
- Precautionary: Coke Oven Battery 4, CO Gas Main Corridor, Battery 4 Platform P2

6. Counterfactual — recorded event vs. with KAVACH

In the recorded composite the incident occurred at 10:30. Actioning this plan at the 06:30 compound alert would have withdrawn the crew 240 minutes (4.0 h) earlier, and pre-empted 2 downstream escalation(s) that deepened the hazard.

Downstream escalations a respected suspension would have prevented:

- **08:30** (t=390): HW-2101 — Hot Work in platform_p2
- **09:30** (t=450): WO-8815 — corrective in gm_corridor

7. Regulatory basis — sources

Compliance coverage: **13 clauses** across **3 sources**, spanning rules R1, R2, R3, R4, R5, R6, R7.

- **DGMS** — DGMS Guidance — Gas Testing, Continuous Monitoring & Confined Working (Mines Act 1952 framework; Coal Mines Regulations (CMR) 2017; DGMS (Tech.) Circulars). Directorate General of Mines Safety (DGMS), Ministry of Labour & Employment, Government of India.
- **FA-1948** — The Factories Act, 1948 (Act No. 63 of 1948, as amended). Government of India (Ministry of Labour & Employment); enforced by State Factory Inspectorates / DISH.
- **OISD-STD-105** — OISD-STD-105 — Work Permit System (Revision I, September 2004). Oil Industry Safety Directorate (OISD), Ministry of Petroleum & Natural Gas, Government of India.

Regulatory texts are faithful summaries of public provisions, provided for demonstration; verify against the official source before operational use.

8. Methodology & data statement

KAVACH fuses signal shape (trend, sub-threshold drift, cross-channel corroboration), work context (permits, work orders, shift handovers), plant topology (zone connectivity) and institutional memory (a single-point gas test, an omitted handover) into eight compound rules (R1–R8) that score each zone 0–100 with hysteresis. The single-sensor baseline and all metrics (lead time, false negatives / positives) are computed against ground truth by the same deterministic engine that produced this report.